IIT Kharagpur ML Security Hackathon - Problem 1
Conditional HPC Anomaly Detection for Backdoor Inference

Objective

Learn normal inference behavior from clean HPC traces and flag deviations. The supplied data has 800 clean inferences and no attack examples.

Data

Counters: cycles, cache-references, LLC-loads. The unnamed sequential column is a row index and is dropped. Cache references and LLC loads have approximately 0.92 Pearson correlation in the supplied clean traces.

Proposed method

A full-covariance Gaussian mixture models the joint distribution of the three log-transformed counters. For each counter, calculate its conditional negative log likelihood given the other two counters. Average the three conditional scores for the final anomaly score. This pseudo-likelihood measures whether each counter is consistent with the clean relationships learned from the other counters. Select mixture size by BIC on fitting data. Calibrate the final threshold on separate clean traces.

Evaluation

25 deterministic clean-only splits; each uses 60% fit, 20% calibration, and 20% held-out clean evaluation. Baselines are a per-counter GMM proxy and a joint GMM on log raw counts.

Mean held-out clean FPR +/- standard deviation
DATE-style per-counter GMM (3-sigma): 3.85% +/- 1.85%
Per-counter GMM proxy at 5% calibration: 5.35% +/- 2.21%
Joint raw-log GMM: 4.60% +/- 1.69%
Conditional GMM pseudo-likelihood: 4.30% +/- 2.08%

Contiguous-block stress check

Eight contiguous blocks of 100 traces were held out in turn. The next block calibrated the threshold; the other six blocks fitted each model.
Conditional pseudo-likelihood clean FPR: 4.50% +/- 3.02%.
DATE-style per-counter GMM (3-sigma): 4.38% +/- 2.62%.
Per-counter GMM proxy at 5% calibration: 4.62% +/- 2.67%.
Joint raw-log GMM: 4.88% +/- 4.12%.
No method has a demonstrated advantage for attack detection, and this single-file experiment cannot establish broad robustness.

Interpretation and limits

These results assess clean false alarms only; they do not establish attack detection performance. No backdoor traces were provided, so TPR, attack FPR, F1, and AUROC cannot be calculated. Proxy-trigger runs, if collected, must be reported separately and kept out of fitting and calibration. Calibration assumes future clean traces are exchangeable with calibration

traces; it does not guarantee the same false-positive rate under shift.
The conditional scoring design differs from the per-counter GMM design in the closest published HPC backdoor-detection paper, but this is not a claim that conditional likelihood or GMMs are new in general.

Related work

Alam, Wang, Maniatakos, DATE 2024, Detecting Backdoor Attacks in Black-Box Neural Networks through Hardware Performance Counters.
https://past.date-conference.com/proceedings-archive/2024/DATA/1142_pdf_upload.pdf

Reproduce: pip install -r requirements.txt

```
python src/train_detector.py --input data/clean_hpc.csv --output artifacts
python src/evaluate_clean.py --input data/clean_hpc.csv --output artifacts
python src/evaluate_blocked.py --input data/clean_hpc.csv --output artifacts
```